

Continual Learning and Model Editing for Foundation Models: A Comprehensive Survey

Abstract

The advent of Foundation Models (FMs) has fundamentally shifted the paradigm of artificial intelligence, enabling unprecedented capabilities across language, vision, and multimodal domains. However, the static nature of pre-trained weights poses significant challenges in dynamic environments where knowledge evolves, tasks accumulate, and safety requirements change. This survey provides a comprehensive review of two critical research directions addressing these challenges: Continual Learning (CL) and Model Editing (ME) for Foundation Models. We synthesize evidence from 300 recent academic studies to construct a unified taxonomy of methodologies, ranging from parameter-efficient fine-tuning and sparse updates to locate-then-edit frameworks and memory-augmented architectures. We critically analyze the trade-offs between stability and plasticity, the mechanisms of catastrophic forgetting, and the scalability of editing operations. Furthermore, we examine emerging frontiers including lifelong editing, cross-modal adaptation, unlearning, and the security implications of model manipulation. By integrating theoretical insights with empirical evaluations across diverse domains such as healthcare, robotics, and time-series forecasting, this work identifies key bottlenecks and outlines a roadmap for developing robust, adaptable, and safe foundation models.

1. Introduction

Foundation Models (FMs), characterized by their massive scale and pre-training on vast corpora, have become the cornerstone of modern artificial intelligence [1]. Despite their remarkable zero-shot and few-shot capabilities, FMs face inherent limitations when deployed in non-stationary environments. The primary challenge is **catastrophic forgetting**, where adapting a model to new tasks or data distributions results in the degradation of previously acquired knowledge [2][3]. Traditional fine-tuning approaches often disrupt the delicate weight configurations established during pre-training, leading to a collapse in general performance [4][5]. Consequently, the field has pivoted towards **Continual Learning (CL)** strategies that enable sequential adaptation while preserving prior competencies, and **Model Editing (ME)** techniques that allow for precise, localized updates to specific facts or behaviors without full retraining [6][7].

The evolution of CL for FMs has moved beyond simple replay buffers and regularization techniques. Recent advances emphasize **parameter-efficient tuning** (e.g., LoRA, adapters) and **sparse updates** that isolate task-specific knowledge, thereby minimizing interference [8][9]. Simultaneously, Model Editing has emerged as a distinct discipline focused on “surgical” modifications. Early methods like ROME and MEMIT utilized causal tracing to identify and modify specific feed-forward network (FFN) layers acting as key-value memories [10][11]. However, scaling these methods to lifelong sequences of edits remains a formidable challenge, often resulting in model collapse or locality violations [12][13].

This survey synthesizes the current state of the art by categorizing methods into coherent families based on their underlying mechanisms: **Parameter Isolation and Sparse Updates**, **Locate-Then-Edit and Causal Intervention**, **Memory-Augmented and Retrieval-Based Systems**, and **Geometric and Optimization-Constrained Approaches**. We draw upon a diverse set of 300 papers to illustrate how these techniques are applied across modalities, including text [14], vision [15], audio [16], and time-series data [2][17]. We also address critical issues such as

safety and unlearning [18][19], **multilingual and cross-modal transfer** [20][21], and the **theoretical limits** of editing [22]. By providing a rigorous analysis of evaluation metrics, scalability bottlenecks, and deployment realities, this work aims to guide future research toward building foundation models that are not only powerful but also adaptable, reliable, and safe throughout their lifecycle.

2. Method

The methodological landscape for adapting Foundation Models is vast and rapidly evolving. We organize the field into six primary taxonomic categories: (1) Parameter-Efficient and Sparse Adaptation, (2) Locate-Then-Edit and Causal Intervention, (3) Memory-Augmented and Retrieval-Based Editing, (4) Geometric and Optimization-Constrained Methods, (5) Lifelong and Sequential Editing Frameworks, and (6) Domain-Specific and Multimodal Adaptation.

2.1 Parameter-Efficient and Sparse Adaptation

A dominant strategy for continual learning in FMs involves restricting updates to a small subset of parameters or auxiliary modules, thereby preserving the integrity of the pre-trained backbone.

2.1.1 Sparse Parameter Updates and Importance Estimation Selective updating of parameters based on their importance to specific tasks has proven effective in mitigating forgetting. **PIECE** (Parameter Importance Estimation-based Continual Enhancement) computes importance scores via Fisher Information or second-order normalization, updating only the top 0.1% of critical parameters identified by curvature analysis [8]. This approach significantly outperforms full fine-tuning and standard replay methods on benchmarks like TRACE. Similarly, **Selective Parameter Update (SPU)** localizes updates to the first MLP layer of transformer blocks using gradient-based scoring, demonstrating that foundation models exhibit implicit modularity where specific parameters encode task-relevant information [9].

Neuron-level sparsity offers another avenue for precision. **NMKE** (Neuron-Specific Masked Knowledge Editing) distinguishes between knowledge-general and knowledge-specific neurons via entropy-guided dynamic masking, preventing interference with general capabilities during lifelong editing [23]. **FiNE** (Fine-grained Neuron-level Knowledge Editing) further refines this by computing contribution scores for individual FFN neurons, excluding final layers to preserve linguistic coherence while achieving superior locality [24]. In the context of unlearning, **TARS** (Targeted Angular Reversal) constructs concept-targeting vectors to reverse the angular alignment of specific feed-forward weights, effectively eliminating bidirectional knowledge access with minimal edits [18].

2.1.2 Low-Rank Adaptation and Adapter Merging Low-Rank Adaptation (LoRA) and adapter-based methods have become standard for efficient continual learning. **Incremental LoRA Merging** applies per-task low-rank matrices to pretrained backbones, merging them directly into weights after each task. Empirical studies reveal that lower LoRA ranks induce less forgetting in Vision Transformers (ViTs) while maintaining plasticity, although ResNets exhibit higher forgetting rates [3]. To address interference in sequential tasks, **STABLE** (Gated Continual Self Editing) integrates a gating layer into the self-editing loop to validate LoRA adapter merges, using metrics like Exact Match drop or KL divergence to enforce stability budgets [25].

Advanced merging strategies explore the geometry of weight spaces. **Task Arithmetic** defines task vectors as the difference between fine-tuned and pre-trained weights, enabling operations like nega-

tion for unlearning or addition for multi-tasking, provided the tasks are approximately orthogonal [26]. **Merge & Initialization Strategies** for diffusion models evaluate orthogonal initialization versus standard reinitialization, finding that standard reinitialization often balances plasticity and stability better than orthogonal constraints which can suffer from task vector alignment issues [27]. **PORTLLM** extends this by proving that personalization adapters possess significantly lower rank than pretraining updates, allowing training-free transfer of LoRA patches to evolved base models with negligible residual error [28].

2.1.3 Feature Space Regularization Rather than constraining parameters, some methods regularize the feature space to preserve pre-trained representations. **LDIFS** minimizes the L2 distance between intermediate feature representations of pre-trained and fine-tuned models, showing that feature-space vicinity correlates more strongly with preserved downstream performance than parameter-space constraints [4]. **Proxy-FDA** aligns local structures of feature distributions via nearest neighbor graph transfer, constructing positive/negative sets based on pre-trained features to mitigate forgetting more effectively than simple distance minimization [29]. **VeFA** (Vector-based Feature Space Adaptation) freezes pre-trained weights and applies element-wise diagonal scaling to input features, strictly constraining updates within the column space of the original weights to eliminate “intruder dimensions” that cause forgetting [30].

2.2 Locate-Then-Edit and Causal Intervention

This family of methods relies on the hypothesis that factual knowledge is stored in specific, localized modules within the transformer architecture, typically within the Feed-Forward Networks (FFNs).

2.2.1 Causal Tracing and Rank-One Updates The seminal **ROME** (Rank-One Model Editing) method identifies mid-layer MLP modules as key-value memories via causal tracing and computes a rank-one update to insert new factual associations [10]. While effective for single edits, ROME struggles with scalability. **MEMIT** (Mass-Editing Memory in a Transformer) extends this by distributing residual updates across a range of critical layers, enabling the simultaneous editing of thousands of facts [11]. However, sequential application of these methods often leads to catastrophic forgetting due to layer weight drift [13].

Recent innovations refine the localization process. **LGA** (Layer Gradient Analysis) identifies “golden layers” via first-order gradient attribution, achieving near-optimal performance with fixed layer selection across diverse queries [31]. **MEMITCSK** varies edit token locations (subject, verb, object) and optimizes layer selection via moving averages of Average Indirect Effect, revealing that commonsense plausibility correlates with early MLP layers [32]. **Does Knowledge Localization Hold True?** challenges the entity-centric view, demonstrating that relational knowledge distributes significantly across attention modules and higher MLP layers, suggesting that relation-centric editing requires modifying attention parameters often ignored by standard methods [33].

2.2.2 Neuron-Level and Subspace Editing Moving beyond layer-level edits, neuron-specific interventions offer finer control. **KNE** (Knowledge Neuronal Ensemble) aggregates related knowledge neurons into ensembles to mitigate localization coupling, employing token-level gradient attribution for precise selection [34]. **NSE** (Neuron-level Sequential Editing) employs weight rewinding to compute target hidden states using original parameters, preventing cumulative drift during iterative multi-layer editing [35]. **SCALPEL** trains low-rank LoRA adapters to minimize probability

gaps on target tasks while maximizing text regularization, operating on the insight that capabilities occupy low-dimensional subspaces rather than discrete modules [36].

Subspace editing techniques leverage linear algebra to isolate specific behaviors. **HulluEdit** decomposes hidden states into orthogonal visual evidence, anti-prior, and residual subspaces via weighted SVD, enabling selective suppression of hallucinatory patterns in Large Vision-Language Models (LVLMs) without interfering with visual grounding [37]. **ZeroUnlearn** reformulates unlearning as a null-space projection, projecting sensitive input representations onto a space orthogonal to original embeddings using SVD-derived matrices [38]. **Gabliteration** extracts top-k refusal directions via multi-directional SVD and applies ridge-regularized projection to alter behavioral subspaces, capturing higher-dimensional patterns beyond single axes [39].

2.3 Memory-Augmented and Retrieval-Based Systems

To decouple knowledge storage from model parameters, memory-augmented approaches treat edits as external retrievable items or distinct memory modules.

2.3.1 External Memory and Retrieval-Augmented Generation SERAC (Semi-Parametric Editing with a Retrieval-Augmented Counterfactual Model) wraps a black-box base model with an explicit edit cache, a scope classifier, and a counterfactual model. This decoupling allows for high reliability and locality, though it struggles with portability to rephrased queries [40]. **EREN** (Edit models by REading Notes) caches edits as natural text notes in an external notebook, utilizing a dual-encoder retriever and instruction-tuned LLM to perform reading comprehension, effectively reformulating editing as a retrieval task [41]. **MeLLO** maintains edited facts in an external retrieval index and decomposes multi-hop queries into sequential sub-questions, significantly outperforming parameter-update methods on multi-hop reasoning tasks where the latter fail catastrophically [42].

MindBridge decouples knowledge storage by encoding edits as a standalone “memory modality,” pre-training a BERT-based encoder to embed factual triplets. This allows reusable editing modules across different LLMs, inspired by low-coupling in multi-modal systems [43]. **PlugLM** replaces FFN layers with Differentiable Plug-in Memory (DPM), storing knowledge as explicit natural language triplets mapped to key-value vectors, enabling training-free knowledge updates [44].

2.3.2 Parametric Memory and Dual-Memory Architectures Internal parametric memory structures offer a middle ground. **WISE** implements a dual parametric memory scheme: a main memory retaining pretrained knowledge and a side memory storing edits in FFN value matrices. It utilizes knowledge sharding with random gradient masks to prevent catastrophic forgetting during lifelong editing [45]. **Larimar** integrates a fixed-size distributed episodic memory matrix with a frozen LLM decoder, updating the memory posterior in one-shot by solving least-squares linear systems, achieving editing speeds 4-10x faster than ROME [46]. **CoachHook** decouples editing weights from base parameters via fixed-size hook layers attached to FFN projection layers, preventing linear memory growth in sequential editing [47].

2.4 Geometric and Optimization-Constrained Methods

Geometric perspectives on weight spaces and optimization landscapes have led to robust editing algorithms that respect the structural properties of the model.

2.4.1 Orthogonal and Null-Space Constraints Preserving the geometry of the weight matrix is crucial for numerical stability. **MOSE** (Multiplicative Orthogonal Sequential Editing) replaces additive weight updates with left-multiplication by an orthogonal matrix derived via SVD, strictly preserving matrix norms and condition numbers to prevent instability accumulation [48]. **X-Edit** applies null-space projection to medical Vision Transformers, computing a closed-form update constrained to the null space of anchor sample representations to prevent interference with prior knowledge [49]. **BETAEDIT** reintroduces a pre-trained knowledge preservation term to penalize approximate null-space leakage, aggregating past edit directions to implicitly reduce cumulative weight drift [50].

RoSE (Robust Same-subject Editing) addresses the ‘‘Covariance Trap’’ by replacing anisotropic covariance matrices with isotropic identity matrices, suppressing activation deviation caused by prompt variations in same-subject editing scenarios [51]. **DG-Hard** applies the Donoho-Gavish optimal hard singular-value threshold to fine-tuning weight deltas, filtering out noise residuals while retaining high-energy task signals to recover damaged capabilities post-hoc [52].

2.4.2 Information-Theoretic and Regularization Approaches Information theory provides a framework for balancing compression and sufficiency during edits. **IBKE** (Information Bottleneck Knowledge Editor) formulates editing as an information-constrained optimization problem, compressing gradient signals into compact latent representations to isolate essential features for generalization [53]. **CORSA** integrates Sharpness-Aware Minimization (SAM) to flatten the loss landscape, employing a composite objective that maximizes the log-likelihood margin between new and old knowledge to enhance adapter stability [54]. **FINCH** derives a theoretical bound linking per-step forgetting to learning rate and loss magnitude, implementing an inverse-loss adaptive schedule to preserve high-loss tokens during fine-tuning [55].

LyapLock reformulates sequential editing as a constrained long-term optimization problem, introducing virtual queues to transform cumulative preservation loss bounds into queue stability problems solvable via Lyapunov optimization [56]. **EtCon** (Edit then Consolidate) employs a two-stage process: Targeted Proximal SFT restricted to FFN layers followed by Group Relative Policy Optimization (GRPO) to consolidate edits at the trajectory level, ensuring reliable generation under autoregressive conditions [57].

2.5 Lifelong and Sequential Editing Frameworks

Scaling editing to lifelong sequences requires mechanisms to handle distribution shifts and error propagation over time.

2.5.1 Normalization and Statistical Tracking **ULTRAEDIT** introduces lifelong normalization, dynamically updating running mean and variance statistics of internal feature representations across editing turns to prevent distributional shift and edit collapse [58]. **STABLEEDIT** extends this with an explicit warm-up phase and full covariance whitening, forming a self-reinforcing stability loop where recursive Bayesian tracking reduces estimation error over time [59]. **D4S** (Dump for Sequence) accumulates editing history statistics into constant-space covariance matrices, theoretically linking aggregated inverse covariance to suppressed norm growth and performance decline [60].

2.5.2 Dynamic Architecture and Expert Expansion Dynamic expansion allows models to grow capacity for new knowledge. **LEM oE** aligns Mixture of Experts (MoE) networks with

sequential data batches, freezing prior experts during new edits and implementing KV anchor routing to guarantee deterministic expert selection [61]. **Transformer-Patcher** freezes original parameters and appends a single trainable neuron to the last FFN layer per mistake, decoupling new knowledge from original parameters to prevent interference [62]. **Incremental Rank Update (IRU)** represents layer weights as linear combinations of low-rank factors, freezing old factors and adding trainable rank-1 updates for new tasks to achieve zero-forgetting [63].

2.5.3 Sequential Editing Challenges Despite advances, sequential editing remains fragile. **Model Editing at Scale leads to Gradual and Catastrophic Forgetting** identifies a two-phase forgetting process: gradual proficiency loss followed by abrupt collapse triggered by disabling edits that break inter-layer compatibility [13]. **Navigating the Dual Facets** systematically assesses that parameter-modifying methods degrade general capabilities after few sequential edits due to parameter divergence, whereas parameter-preserving methods struggle with generalization to rephrased queries [64]. **WilKE** (Wise-Layer Knowledge Editor) attributes performance collapse to “pattern mismatch,” where predefined editing layers do not align with the specific layers detecting input patterns, proposing dynamic layer selection to mitigate this [65].

2.6 Domain-Specific and Multimodal Adaptation

Adaptation techniques are increasingly tailored to specific domains and modalities, addressing unique challenges in time-series, healthcare, robotics, and code.

2.6.1 Time-Series and Sensor Data Time-series foundation models exhibit distinct forgetting patterns. **Are Time Series Foundation Models Susceptible to Catastrophic Forgetting?** demonstrates that high learning rates drive rapid adaptation but induce severe forgetting, while low rates hinder new pattern acquisition [2]. **VisionTS++** reformulates time-series forecasting as image reconstruction via continual pre-training of Masked Autoencoders (MAE), leveraging vision-model filtering to discard samples violating pixel range constraints [66]. **GlucoFM** employs a dual-stream state-event decomposition for continuous glucose monitoring, aligning irregular traces to a chronological grid to capture complementary clinical signals [67]. **TimeBlocks** constructs lightweight models at inference by sequentially selecting pre-trained processing blocks from a clustered “Blockbase,” enabling dynamic assembly of specialized architectures [68].

2.6.2 Medical and Healthcare Applications In healthcare, privacy and domain shift are paramount. **UNICON** employs frozen image and text encoders with incremental adaptation modules (LoRA + Adapters) for medical foundation models, preventing catastrophic forgetting during multi-step adaptation across tasks and anatomical regions [69]. **COMFORT** pre-trains foundation models exclusively on healthy individual data to bypass patient privacy constraints, utilizing library-based storage of low-rank adapters for downstream adaptation [70]. **MedEditBench** introduces a rigorous evaluation framework for medical knowledge editing, demonstrating that editing with self-generated rationales (SGR-Edit) significantly outperforms answer-only editing by embedding underlying logical structures [71].

2.6.3 Multimodal and Vision-Language Models Editing multimodal models requires handling cross-modal dependencies. **MMEdit** benchmarks MLLM editing, revealing that editing language modules yields higher reliability than vision modules, and memory-based methods often sacrifice multimodal locality [72]. **BalancedEdit** wraps transformer layers with modules containing discrete keys and dynamic influence radii, approximating fact-specific influence scopes via latent

space distances to balance generality and locality in multi-modal editing [73]. **SmartEdit** integrates MLLM hidden states via bidirectional interaction modules to enable complex instruction-based image editing, highlighting that direct integration is insufficient without explicit feature interaction [74]. **HIRE** (Hallucination-aware Intermediate Representation Edit) employs a dual-encoder autoencoder to disentangle semantic and hallucinatory subspaces in LVLMs, projecting features onto semantic manifolds to mitigate hallucinations [75].

2.6.4 Code and Software Engineering Code editing involves structural constraints and version control. **Grace** frames code editing as an infilling task conditioned on prior “associated edits,” significantly boosting exact match accuracy by capturing developer intent [76]. **Proc3D** utilizes a Procedural Compact Graph (PCG) representation for 3D shape generation, enabling real-time parametric editing by abstracting procedural operations into a compact graph [77]. **EDITCODER** fine-tunes code LLMs on curated GitHub commits, demonstrating that models pre-trained on commit data outperform those trained on general code, though evaluation requires reproducing entire code segments [78].

3. Challenges and Outlook

Despite significant progress, several critical challenges hinder the widespread deployment of continual learning and model editing for Foundation Models.

3.1 Scalability and Sequential Stability

The most pressing limitation is the inability of current methods to scale to lifelong sequences of edits without degradation. Parameter-modifying methods like ROME and MEMIT exhibit catastrophic forgetting after approximately 100-1000 edits due to weight drift and inter-layer incompatibility [13][22]. While memory-based methods like SERAC offer better stability, they struggle with portability and inference latency [40][79]. The “Covariance Trap” and dimensional collapse in representation spaces create fragile directions where small perturbations amplify disproportionately, limiting the theoretical capacity for sequential updates [51][22]. Future research must focus on **geometry-aware merging** and **dynamic architecture expansion** that can accommodate infinite edit streams without performance collapse.

3.2 Evaluation Bottlenecks and Benchmarking

The field suffers from a lack of unified benchmarks and standardized metrics. Many studies rely on static datasets like ZsRE or CounterFact, which fail to capture the complexity of real-world sequential updates [12][80]. Recent efforts like **OAKS** benchmark online adaptation to continual knowledge streams, revealing significant state-tracking failures in current models [81]. **MLaKE** and **VLKEB** highlight the poor cross-lingual and multi-hop reasoning capabilities of existing editing methods [21][82]. There is an urgent need for **dynamic, lifelong benchmarks** that evaluate not just single-edit success but also long-term retention, reasoning consistency, and computational efficiency over thousands of updates.

3.3 Safety, Security, and Unlearning

Model editing poses serious safety risks. **Position: Editing Large Language Models Poses Serious Safety Risks** categorizes malicious use cases including backdoor insertion, bias injection, and jailbreaking, noting that edited models are often indistinguishable from benign ones [83].

BadEdit demonstrates that backdoor injection can be reframed as a lightweight knowledge editing problem, achieving 100% attack success rates [84]. Conversely, **unlearning** remains imperfect; methods like **TARS** and **ZeroUnlearn** achieve targeted erasure but may degrade model utility or fail to remove out-of-distribution knowledge [18][38]. **Grokking-Enhanced Unlearning** suggests that post-grokking states offer better gradient orthogonality for selective forgetting, but this requires extended training [85]. Robust **detection mechanisms** like **KETI** are needed to identify malicious edits, alongside **provably safe unlearning** algorithms that guarantee data removal without utility loss.

3.4 Multimodal and Cross-Domain Generalization

Current editing methods are predominantly text-centric. Editing vision, audio, and multimodal models introduces complexities in **cross-modal alignment** and **localization**. **SAKE** reveals that current methods struggle to generalize edits to equivalent auditory inputs while preserving intra-attribute knowledge [86]. **Cross-Lingual Knowledge Editing** studies show that language modeling gaps significantly influence editing efficiency, with English editing often outperforming other languages due to pre-training data disparities [87]. Future work must develop **modality-agnostic editing frameworks** that can handle heterogeneous data types and ensure consistent behavior across languages and sensory inputs.

3.5 Deployment and Efficiency

Deployment constraints, particularly in edge and real-time settings, remain a barrier. **Energy-aware Incremental OTA Update** highlights the energy costs of firmware updates in batteryless IoT devices, necessitating deferred write pipelines [88]. **Real-time Adapting Routing (RAR)** proposes shadow inference mechanisms to balance cost and quality in layered foundation models, but sequential request ordering affects guide memory population [89]. **Speculative Decoding** offers speedups for time-series models but saturates with large block sizes [90]. Efficient **inference-time interventions** and **training-free adaptation** methods like **PORTLLM** and **Activation Steering** [91][28] offer promising directions for low-latency deployment, but their scalability to massive models and complex tasks requires further validation.

4. Conclusion

This survey has synthesized the rapidly evolving landscape of Continual Learning and Model Editing for Foundation Models. We have categorized methodologies into parameter-efficient adaptation, causal intervention, memory augmentation, geometric constraints, and lifelong frameworks, drawing on evidence from 300 distinct studies. While significant strides have been made in isolating knowledge updates and mitigating catastrophic forgetting, fundamental challenges in scalability, safety, and multimodal generalization persist. The transition from single-edit proofs-of-concept to robust, lifelong adaptation systems requires a paradigm shift towards **geometry-aware optimization**, **dynamic architectural growth**, and **rigorous, dynamic benchmarking**. As Foundation Models become increasingly integrated into critical infrastructure, developing editing and continual learning techniques that are not only effective but also safe, interpretable, and efficient will be paramount. Future research must bridge the gap between theoretical guarantees and practical deployment, ensuring that foundation models can evolve alongside the dynamic world they serve.

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